DECIMALS, MODULAR CYCLES AND FERMAT ON ONE PAGE

Long division is a walk through remainders

Keep track of what is left over, and a repeating decimal becomes a finite walk.

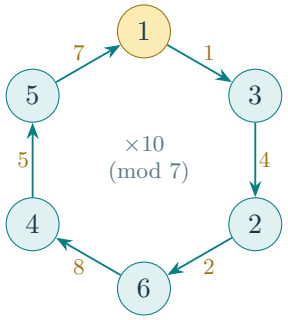
1 / CALCULATE ONE SEVENTH

Record the digit and keep the remainder

Start with $r_0 = 1$. At each step, multiply the remainder by 10, divide by 7, write the digit d_k, and keep the remainder r_k.

k	division step	digit d_k	remainder r_k
1	$10 = 7 \cdot 1 + 3$	**1**	**3**
2	$30 = 7 \cdot 4 + 2$	**4**	**2**
3	$20 = 7 \cdot 2 + 6$	**2**	**6**
4	$60 = 7 \cdot 8 + 4$	**8**	**4**
5	$40 = 7 \cdot 5 + 5$	**5**	**5**
6	$50 = 7 \cdot 7 + 1$	**7**	**1**

Read down the digit column: 142857.



Circles: remainders. Arrow labels: digits.

$$\frac{1}{7} = 0.\overline{142857}$$ Remainder 1 returns: the six steps repeat.

2 / ONLY FINITELY MANY POSSIBILITIES

A finite walk must stop or repeat

With denominator q, the possible remainders are $0, 1, \ldots, q-1$. Each remainder determines the next digit and the next remainder:

$$10r_{k-1} = qd_k + r_k, \qquad 0 \le r_k < q.$$

A zero remainder ends the decimal. Repeating a non-zero remainder forces the same following digits, so the decimal repeats.

A cycle can have a short lead-in:

$$1 \xrightarrow{1} 4 \xrightarrow{6} 4, \qquad \frac{1}{6} = 0.1\overline{6}.$$

For comparison, $1/8 = 0.125$ reaches remainder 0 and terminates.

3 / FOLLOW THE WALK ALGEBRAICALLY

Powers of ten retrace the walk

Writing $x \equiv y \pmod 7$ says that x, y leave the same remainder. Each step gives $r_k \equiv 10r_{k-1} \pmod 7$, so, from $r_0 = 1$,

$$r_k \equiv 10^k \pmod 7.$$

k	0	1	2	3	4	5	6
$10^k \bmod 7$	1	3	2	6	4	5	1

Returning to remainder 1 after L steps means $10^L \equiv 1 \pmod 7$. The first such positive L is the **multiplicative order** of 10 modulo 7:

$$\operatorname{ord}_7(10) = 6.$$

4 / A PRIME FORCES A RETURN

Fermat's little theorem

For prime p and $p \nmid b$, multiplying $1, 2, \ldots, p-1$ by b modulo p rearranges that list: no two distinct entries acquire the same remainder. Taking products gives

$$b^{p-1}(p-1)! \equiv (p-1)! \pmod p.$$

The product $(p-1)! = 1 \cdot 2 \cdots (p-1)$ has no factor p, so it can be cancelled:

$$b^{p-1} \equiv 1 \pmod p$$

With $b = 10$, the theorem applies to every prime except 2 and 5.

5 / COUNT COMPLETE LAPS

The period divides $p - 1$

Here p is prime and $p \neq 2, 5$. The return after $p - 1$ steps completes whole laps, so the first return L satisfies

$$L = \operatorname{ord}_p(10), \qquad L \mid (p-1).$$

p	repeating block	L	$p-1$
7	142857	6	6
11	09	2	10
13	076923	6	12
17	0588235294117647	16	16

Leading zeros are part of the repeating block. Full length $p - 1$ occurs when the walk visits every non-zero remainder.

Change the base. In base $b \ge 2$, use $br_{k-1} = qd_k + r_k$. For prime $p \nmid b$, the period of $1/p$ in base b divides $p - 1$. Here $p \nmid b$ means that p does not divide b.

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